

Article

From Air to Space: A Comprehensive Approach to Optimizing Aboveground Biomass Estimation on UAV-Based Datasets

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Abstract: Estimating aboveground biomass (AGB) is vital for sustainable forest management and helps to understand the contributions of forests to carbon storage and emission goals. In this study, the effectiveness of plot-level AGB estimation using height and crown diameter derived from UAV-LiDAR, calibration of GEDI-L4A AGB and GEDI-L2A rh98 heights, and spectral variables derived from UAV-multispectral and RGB data were assessed. These calibrated AGB and height values and UAV-derived spectral variables were used to fit AGB estimations using a random forest (RF) regression model in Fuling District, China. Using Pearson correlation analysis, we identified 10 of the most important predictor variables in the AGB prediction model, including calibrated GEDI AGB and height, Visible Atmospherically Resistant Index green (VARIG), Red Blue Ratio Index (RBRI), Difference Vegetation Index (DVI), canopy cover (CC), Atmospherically Resistant Vegetation Index (ARVI), Red-Edge Normalized Difference Vegetation Index (NDVI_{re}), Color Index of Vegetation (CIVI), elevation, and slope. The results showed that, in general, the second model based on calibrated AGB and height, Sentinel-2 indices, slope and elevation, and spectral variables from UAV-multispectral and RGB datasets with evaluation metric (for training: $R^2 = 0.941$ Mg/ha, RMSE = 13.514 Mg/ha, MAE = 8.136 Mg/ha) performed better than the first model with AGB prediction. The result was between 23.45 Mg/ha and 301.81 Mg/ha, and the standard error was between 0.14 Mg/ha and 10.18 Mg/ha. This hybrid approach significantly improves AGB prediction accuracy and addresses uncertainties in AGB prediction modeling. The findings provide a robust framework for enhancing forest carbon stock assessment and contribute to global-scale AGB monitoring, advancing methodologies for sustainable forest management and ecological research.



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1. Introduction

Carbon concentration in the atmosphere as a greenhouse gas can influence the global climate pattern. Human-caused activities such as deforestation, forest degradation, industrialization, and forest fires have caused an increase in the atmospheric carbon level and disturbed the global carbon cycle. Nevertheless, nature has developed mechanisms for storing carbon in its natural sinks, and forests play a significant role in the carbon sinks of the terrestrial ecosystem globally [1,2]. It is critical to reduce deforestation and degradation since carbon emissions are increasing over land [3]. Stored carbon in forests is essential

in regulating ecosystem services, and it is measured through five pools: aboveground, below-ground, litter, woody debris, and soil organic matter [1,4]. Aboveground biomass (AGB) is the largest carbon storage pool and can be estimated through destructive and non-destructive methods [5–7].

Remote sensing technologies provide efficient and non-destructive ways of estimating forest AGB at various temporal and spatial scales [6,8]. Integrating passive and active remote sensing datasets and approaches has benefits and limitations [9–11]. The accuracies and limitations are estimated by integrating spaceborne–airborne datasets previously achieved solely from forest inventory datasets [12,13]. In general, spaceborne remote sensing frameworks such as Sentinel-2 and GEDI data can be utilized, which are the most cost-effective for estimating forest AGB on a global scale [14–20].

The uncertainty in AGB estimation of GEDI and other AGB products (ICESat-2 and NISAR) produced due to the dataset's coarser data or unavailability of ground field data can be calibrated using airborne data at the regional level using temporal remote sensing datasets [17,21,22]. The research frameworks can employ various machine learning models, such as random forest, support vector machine, XGBoost, and deep learning algorithms, for extrapolated AGB prediction comparisons using spectral indices and textural features extracted from optical datasets to improve prediction accuracies [6,23–29].

Airborne laser scanners, especially UAV-LiDAR, can provide accurate forest structural information with plot-level to significant area coverage, higher accuracies, lower cost, and stronger adjustability [30–33]. The high spatial resolution of UAV-LiDAR and AGB estimates can be aggregated to coarser resolutions, which bridges the scale gap between field data and other remote sensing datasets [34]. The spectral and textural feature extraction from UAV-RGB datasets is utilized in AGB prediction models in combination with various machine learning algorithms and to develop allometric equations species-wise [35–37]. The vegetation indices extraction from UAV-multispectral datasets can provide vegetation health at the plot level and increase the accuracy of the estimates [38]. Integrating airborne and spaceborne datasets can provide more accurate aboveground biomass estimation at the plot level, which can be utilized for calibrating AGB prediction models [39,40]. This study highlights the integration framework from multi-sensor remote sensing (airborne to spaceborne) datasets to make improved and more complete calibration and estimation assessments.

The traditional forest AGB estimation method involves the destructive approach, which calculates AGB for each tree. The allometric equation is an essential non-destructive method to estimate AGB at the tree-to-plot scale [32,41]. Diameter at breast height (DBH) is a critical predictor for allometric equations for estimating AGB [42,43]. However, DBH cannot be obtained solely from UAV-LiDAR ALS data [44,45]. By contrast, trunk height (H) can be captured by UAV-LiDAR ALS data [46–48], which indicates that it can act as a potential predictor in allometric equations for AGB [49,50]. The crown area can be derived from the UAV-LiDAR data and reflects the size and tree crown shape [45,51]. Therefore, this study is expected to develop a generic allometric equation for estimating AGB using crown diameter and tree height extracted from UAV-LiDAR and compare it with GEDI AGB and height estimates at the plot level, which helps to understand the standard deviation of AGB estimates from airborne to spaceborne datasets. It will further be utilized to calibrate GEDI AGB and height footprints.

Over the last few decades, tropical and sub-tropical forests have suffered depletion due to forest degradation, deforestation, and forest fires [52–55]. It would also impact the forest environment along staggering cliffs near the longest and the largest river in the country, the Yangtze River in China [56]. Fuling district, Chongqing, China, is considered the ecological safety barrier of the Yangtze River and Three Gorges Reservoir Area (TGRA) [57].

Forest AGB degradation and land-use land-cover change have caused ecological and environmental problems in Fuling District [58–60]. Integrating multi-source remote sensing datasets is crucial to identifying forest AGB estimates for the Fuling District.

This study aims to develop a comprehensive methodology and demonstrate the seamless integration of workflow. The main contribution of this study is the proposed integrated method, which combines the following objectives: (i) Obtain plot-level AGB estimation using UAV-LiDAR-derived tree height and crown diameter and calibration factor extraction to calibrate GEDI-L4A and GEDI-L2A (AGB and height) footprints. (ii) Identify and extract the most significant spectral variables from UAV-derived multispectral and RGB datasets contributing to AGB predictions. (iii) Construct random forest regression models utilizing calibrated AGB estimates, slope and elevation from NASADEM, Senitnel-2 indices, and UAV-derived spectral variables to predict AGB within the Fuling District of the Three Gorges Reservoir (TGR) region. The study outcomes will be helpful for local and regional planners to obtain the calibration factor from airborne datasets over the surveyed plots and use it to calibrate the AGB and height footprints. It will be helpful to improve AGB estimates, reduce uncertainties, and improve the carbon stock assessment and forest ecosystem management within Fuling District, China.

2. Materials and Methods

2.1. Study Area

The study area is Fuling District, Chongqing, China, covering the scope along the Yangtze River in the TGR region, as shown in Figure 1, with an area of 2945 km². This district has a sub-tropical humid monsoon climate and abundant sunshine. Spring and summer have rainfall of 1009 mm, with the highest temperature at 35.8 °C occurring in August 2011. In addition, the annual duration of sunshine is 1528 h [57,61,62]. The TGR region is populated with the loftiest mountains and hills, connected by the Yangtze River and the Three Gorges Dam (TGD) reservoir area with sub-tropical evergreen broad-leaf forest (SEBF) vegetation. Citrus plantation cultivation, also of the SEBF type, has become an economic contributor in the region, particularly in the Fuling district [15,20,58,60,63]. There are two main rivers in the Fuling District territory; the first is the Yangtze River and the second is the Wujiang River. These are essential in the TGR region, which has numerous natural protection areas [57]. The dominant species of the Chongqing area are *Pinus massonianna* and *Gordonia acuminata* [64]. The three survey sites within the study area were about 3.75 km² and were all set to conduct a UAV flight survey to acquire UAV-LiDAR, UAV-multispectral, and RGB ortho-photos. This survey aimed to overlap with the GEDI-L4A footprints within the sites and treat all GEDI-L4A footprints as field plots for this study (Figure 1c–e and Table 1). The conceptual framework of this study is mentioned in detail in Figure 2. The summary of key datasets is mentioned in detail in Table 2.

Table 1. UAV survey statistics.

Site Name	Survey Date	Location (DMS)	Surveyed Area (Sq.km)	Number of Field Plots
Site 1	8 October 2024	107°21′52.422″ E, 29°47′19.643″ N	0.74	24
Site 2	2 July 2024	107°28′14.967″ E, 29°42′51.267″ N	1.76	6
Site 3	8 October 2024	107°29′13.703″ E, 29°40′34.452″ N	1.25	30

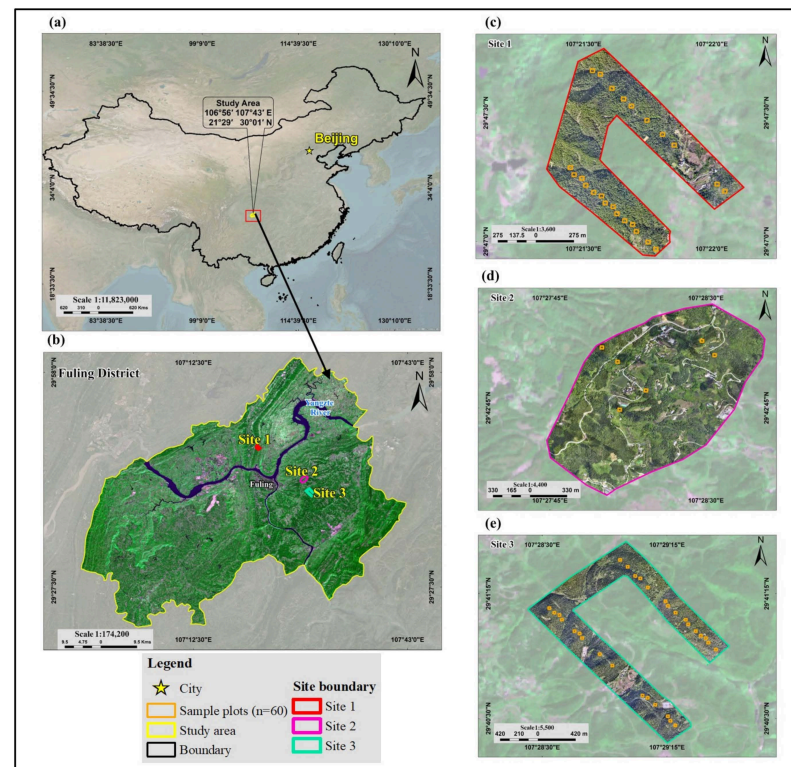


Figure 1. Study area description. (a) Location of Fuling District, Chongqing in China, (b) a Sentinel-2 composite image acquired from 1 August to 20 November 2024, covering the study area (Fuling District) (yellow outline), (c) a UAV image acquired on 2 July 2024, covering the Site 1 (red outline), and the field plots (orange outline), (d) UAV data acquired on 8 October 2024, covering Site 2 (purple outline), and (e) UAV data acquired on 8 October 2024, covering Site 3 (turquoise green outline).

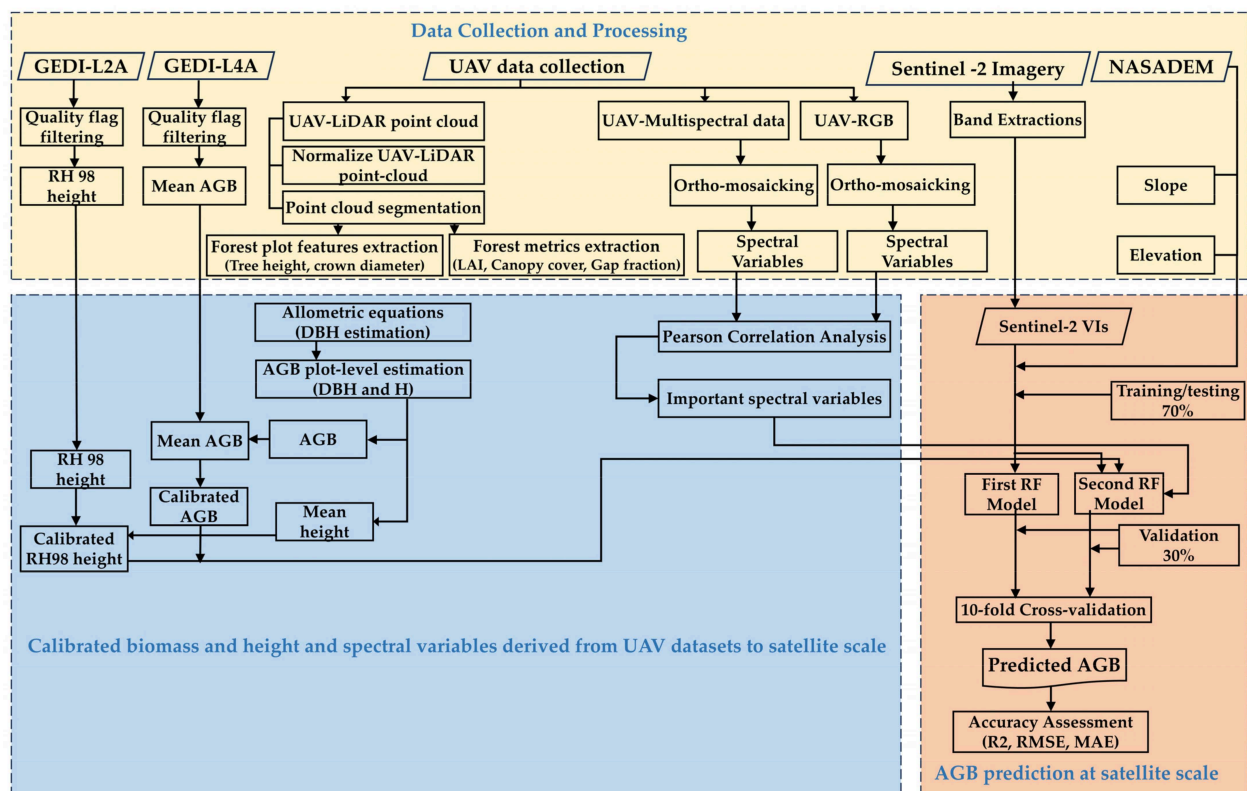


Figure 2. Conceptual framework of the study.

Table 2. Summary of key datasets in the study.

Name	Resolution	Role	Source
GEDI-L4A	25 m	Provides the mean aboveground biomass (AGB) value	GEE Image Collection “LARSE/GEDI/GEDI04_A_002_MONTHLY”
GEDI-L2A-rh98	25 m	Measures the height value (canopy height)	GEE Image Collection “LARSE/GEDI/GEDI02_A_002_MONTHLY”
Sentinel-2	10 m	Provides spectral bands to derive spectral indices	GEE Image Collection “COPERNICUS/S2_SR”
Slope and Elevation (NASADEM)	10 m	Extracts slope and elevation data	GEE Image Collection “NASA/NASADEM_HGT/001”
UAV-LiDAR	Point Cloud	Captures height and crown diameter data for estimating plot-level AGB	Self-Acquired Platform
UAV-Multispectral and RGB imagery	0.12 m	Extracts high-resolution spectral variables for detailed analysis	Self-Acquired Platform

2.2. Dataset

2.2.1. GEDI-L4A Data Collection and Processing

The spaceborne LiDAR resource (GEDI) offers precise forest structural information and derived biomass estimates [19,65]. The GEDI-L4A raster AGB version 2.1 dataset [66] was downloaded from the Google Earth Engine (GEE) data repository between 25 March 2019 and 1 March 2023. The datasets provide a 25 m resolution of mean aboveground biomass density (AGBD) in Mg/ha for each geolocated footprint. The product represents footprint-level AGBD derived using parametric linear models relating GEDI-L2A waveform relative height (RH) metrics with AGBD [23]. GEDI-L4A accuracy has been evaluated by directly associating GEDI-L4A estimates at 25 m ground data [67,68]. The GEDI-L4A product, including the AGB density pre-processing, was coded in GEE. The timeline for the AGB density for GEDI-L4A was limited to 1 January 2022 to 31 December 2022. These datasets contain a set of quality flags that allow us to remove low-intensity and potential atmospheric problems [19]. We applied quality filtering to have L2A and L4A quality flags equal to 1, indicating the removal of low-sensitivity footprints. We also applied algorithm flags equal to 1 and sensitivity greater than 0.95 to ensure an adequate signal-to-noise ratio (SNR), and a degraded flag equal to 0 to remove the degraded pointing state or positioning information. Moreover, we removed those footprints by dividing agbd_SE by agbd and showed a ratio value of less than 50. The slope mask was also applied to the footprints, selecting those with slopes less than 30 m [21,69]. The total GEDI-L4A footprints exported were 10,345. After data filtering, the GEDI-L4A raster datasets were exported and converted into polygons with 25×25 m footprints. All footprints were further refined by clipping with the developed forest layer [20] and re-checking on high-resolution imagery to ensure every footprint was over the forest class. The final GEDI-L4A AGB footprints, after all quality checks in the Fuling District, total 2096.

2.2.2. GEDI-L2A Product Processing

The GEDI-L2A geolocated elevation and height metrics product version 2 is the raster version of the original product [70,71] and was downloaded from the Google Earth Engine (GEE) data repository available from 25 March 2019 until 1 March 2023 with a 25 m resolution of rh98 relative heights (RH) metrics. The timeline of the relative height footprints for GEDI-L2A was the same from 25 March 2019 to 1 March 2023 to overlap with the GEDI-L4A footprints for further analysis. The quality flag, degraded flag, and urban proportion were used as the quality-filtering criteria [70]. Furthermore, GEDI-L2A raster datasets were exported and converted into polygons with 25×25 m footprints. All footprints of GEDI L2A were quality filtered by aligning with the selected footprints GEDI

L4A mentioned in the above 2.2.1 section. The final GEDI-L2A points, including height metrics rh98 in the Fuling District, total 2096.

2.2.3. Sentinel-2 and NASADEM Data Processing

The Sentinel-2 multispectral imagery is freely available over the GEE platform provided by the European Space Agency (ESA) [24,26,72,73]. The satellite data has 13 spectral bands with several spatial resolutions, which is significant for vegetation analysis [74,75]. Sentinel-2A data was from “COPERNICUS/S2_SR” in Google Earth Engines, and a code editor was used to process and download the image. The timelines were from 1 August 2024 to 20 November 2024, subject to cloud masking and temporal and median filtering by the GEE, resulting in a single median composite image. The spectral indices analyses acted as predictor variables in the calibration model of AGB estimation at the GEDI plot level. NASADEM image collection was used to extract the elevation and slope at 30 m resolution over the GEE platform. Its information was also employed in the calibration prediction model of AGB. Overall, 13 spectral variables comprising nine spectral indices and four spectral channels were derived at plot level from both datasets, the details of which are shown in Table 3 [74].

Table 3. Summary of vegetation indices derived from the Sentinel-2 imagery.

VI	Name	Formula
ARVI	Atmospherically Resistant Vegetation Index	$(\rho\text{NIR} - (2\rho\text{Red} - \rho\text{Blue})) / (\rho\text{NIR} + (2\rho\text{Red} - \rho\text{Blue}))$
CIg	Green Chlorophyll Index	$\rho\text{NIR} / \rho\text{Green} - 1$
DVI	Difference Vegetation Index	$\rho\text{NIR} - \rho\text{Red}$
EVI	Enhanced Vegetation Index	$2.5(\rho\text{NIR} - \rho\text{Red}) / (\rho\text{NIR} + 6\rho\text{Red} - 7.5\rho\text{Blue} + 1)$
GNDVI	Green Normalized Difference Vegetation Index	$(\rho\text{NIR} - \rho\text{Green}) / (\rho\text{NIR} + \rho\text{Green})$
MSAVI	Modified Soil Adjusted Vegetation Index	$[2\rho\text{NIR} + 1 - \sqrt{(2\rho\text{NIR} + 1)^2 - 8(\rho\text{NIR} - \rho\text{Red})}] / 2$
NDVI	Normalized Difference Vegetation Index	$(\rho\text{NIR} - \rho\text{Red}) / (\rho\text{NIR} + \rho\text{Red})$
SR	Simple Ratio	$\rho\text{NIR} / \rho\text{Red}$
VARIg	Visible Atmospherically Resistant Index green	$(\rho\text{Green} - \rho\text{Red}) / (\rho\text{Green} + \rho\text{Red} - \rho\text{Blue})$

2.2.4. UAV-LiDAR Data Processing

The UAV-LiDAR raw point clouds were acquired on July 02 and October 8, 2024, covering 3.75 km² on three sites (Figure 1c–e). They were acquired using a sensor DJI L1 over a platform DJI M300RTK with a flight altitude of 120 m, flight speed of 10 m/s, and maximum range of 250 m (Table 4). Keeping the flight conditions (wind, rain, fog, temperature) and sensor calibration (skewed imagery or mismeasurement due to inaccurately calibrated) aspects in mind could lead to inconsistency in the data acquisition. The level 1 (L1) raw LiDAR point clouds data, which are usually in LAS (LASer) or LAZ (compresses LAS) format, were imported into LiDAR processing software LiDAR360 version 5.2 for further processing. The calibration corrections were applied to the data to remove the systematic errors. To ensure that the data are accurate, we referenced a coordinate system using GNSS data. Moreover, noise and erroneous points not representing surface features were filtered out. This involved removing points that deviated significantly from the expected range or spatial distribution. The next step was to apply point cloud segmentation on normalized data derived after subsampling, remove outliers, and classify ground points to obtain the plot-level tree statistics. After that, the forest metrics were calculated using the grid to extract the forest matrix analysis (LAI, canopy cover, gap fraction). The processed point cloud information, such as tree height and crown diameter, over the GEDI plots, was used in plot-wise AGB estimations and the extraction of the calibration factor. The overall workflow is shown in Figure 3.

Table 4. UAV-LiDAR system characteristics and flight parameters for data acquisition.

Setting and Flight Parameters	Description
Maximum scan angle	Repeat Scanning Mode: $70.4^{\circ} \times 4.5^{\circ}$; Non-Repetitive Scanning Mode: $70.4^{\circ} \times 77.2^{\circ}$
Scan Rate/Scan Frequency	160 KHz
Average point density	350 points/m ²
Laser Center wavelength	905 nm
Maximum number of returns	3
Range Accuracy	Plane: $5 \pm \text{cm}$, Elevation: $10 \pm \text{cm}$
Effective measurement rate	240 KHz
UAV and Sensor Weight	7 kg
Digital Camera image resolution	20 megapixels

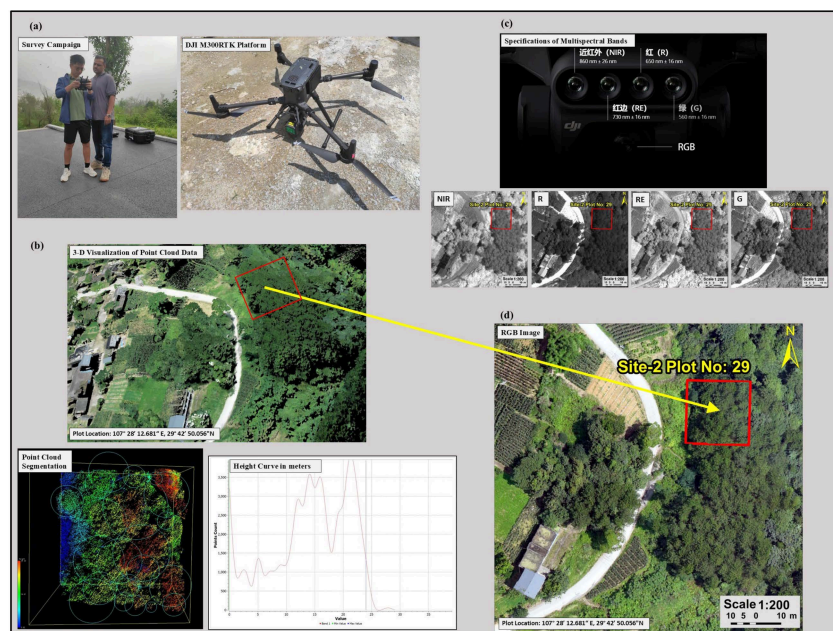


Figure 3. UAV data acquisition and processing workflow. (a) Data acquisition using multi-rotor UAV-LiDAR sensor over real-time kinematic platform, (b) 3D visualization of UAV-LiDAR point cloud data, with point cloud segmentation and height curve of site 2 plot 30 (red outline), (c) a specification of UAV-multispectral and RGB platform showcase for each multispectral band over site 2 plot 30 (red outline), (d) UAV-RGB ortho-mosaic data over site 2 plot 30 (red outline).

2.2.5. UAV-Multispectral and UAV-RGB Data Processing and Their Spectral Variables

Multispectral images were acquired on 2 July 2024 and 8 October 2024, covering 3.75 km², using a multirotor DJI Mavic 3M, including four multispectral cameras (green, red, red edge, and near-infrared) and an RGB camera. Keeping the flight conditions and sensor calibration aspects in mind could lead to inconsistency in the data acquisition. The UAV was equipped with a high-performance GNSS system, a barometer, an accelerometer, a gyroscope, a compass, and a 20-megapixel RGB camera with a lens having an equivalent length of 35 mm and focal mounted on a 3-axis electronic gimbal for stabilization. The GNSS onboard the UAV was complemented with Real-Time Kinematics (GNSS-RTK), which provided a very high horizontal positioning accuracy of ± 1 and vertical positioning accuracy of ± 1.5 cm. Autonomous flights were conducted at a constant velocity of 15 m/s. At the same time, the flight's altitude was kept at 150 m above the ground due to the rugged mountain to maintain consistency in data acquisition in the face of highly varying mountainous terrain. The UAV's imaging system was configured to achieve 80% frontal overlap and 70% lateral overlap, as detailed in Table 5.

Table 5. UAV-multispectral and RGB system characteristics and flight parameters for data acquisition.

Setting and Flight Parameters	Description
Maximum scan angle	61.2° × 48.10°
UAV and Sensor Weight	0.951 kg
Camera image resolution-mega pixels	5 megapixels
RGB Image Size	20 million
Focal Length (mm)	25 mm
Sensor Size (mm)	1/2.8-inch CMOS
Image Size	2592 × 1944

After the data acquisition from the multispectral sensor, data were processed using the software DJI Terra. It performed radiometric calibration to correct for variations in lighting conditions. It also aligned the different spectral bands to the same projection system. The software initially used feature-matching algorithms to align the images and create a sparse point cloud. It then generated the DSM from the dense point cloud, which provides elevation information and helps in accurate image stitching. Afterwards, the images were projected onto the DSM to create a geometrically corrected ortho-mosaic. The spectral bands were then combined to create composite images that reflect different spectral information. The multispectral digital ortho-mosaic was exported at a resolution of 0.1 m, and the RGB ortho-mosaic at 0.05 m. Later, all the multispectral bands and RGB images were clipped to plot boundaries in the sites, as mentioned in Table 1, at the normalized resolution of 0.12 m. The data acquisition and processing workflow for all three sites and all plots are shown in Figure 3.

Some spectral indices are widely used to reflect vegetation properties by merging the surface reflectance of specific band wavelengths. It elaborates on the potential of multispectral and RGB imagery for enhancing the AGB estimation models.

In this study, multispectral imagery comprising four bands (Green, NIR, Red, and Red Edge) and UAV-RGB mosaic imagery were clipped to 60 plot boundaries to standardize the spatial extent for analysis. Individual image clips obtained from the two data types were combined into one raster file, where each raster chip corresponded to one of the 60 plot boundaries. This method ensured that the raster contained all bands and RGB information as input variables for AGB estimation. Moreover, this technique reduced the processing time over the GEE platform since the information was combined into a single format for analysis. This level was employed to extract vegetation indices.

Overall, 39 spectral variables comprising 32 spectral indices and seven spectral channels were derived at plot level from both datasets, the details of which are shown in Tables 6 and 7 [37,39]. Vegetation indices derived from both the multispectral and RGB imagery were merged to improve the accuracy of the prediction in AGB.

Table 6. Summary of color vegetation indices derived from the UAV-multispectral imagery.

VI	Name	Formula
CI _{re}	Chlorophyll Index (red edge)	$CI_{re} = (NIR/R_{re}) - 1$
CVI	Chlorophyll Vegetation Index	$NIR \times R/G^2$
EVI	Enhanced Vegetation Index	$2.5 \times ((NIR - R)/(NIR + 2.4 \times R + 1))$
GNDVI	Generalized Normalized Difference Vegetation Index	$(NIR - G)/(NIR + G)$
CI _{green}	Green Chlorophyll Index	$(NIR/G) - 1$
MSR	Modified Simple Ratio	$(NIR/R) - 1/(\sqrt{(NIR/R)} + 1)$
MSR-re	Modified Simple Ratio (red-edge)	$(NIR/R_{re}) - 1/\sqrt{(NIR/R_{re}) + 1}$
NDVI	Normalized Difference Vegetation Index	$(NIR - R)/(NIR + R)$
NDVI _{re}	Red-Edge Normalized Difference Vegetation Index	$NDVI_{re} = (NIR - R_{re})/(NIR + R_{re})$
RTVI _{core}	Red-Edge Triangular Vegetation Index	$100 \times (NIR - R_{re}) - 10 \times (NIR - G)$
SR _{re}	Simple Ratio (red-edge)	(NIR/R_{re})
GDI	Green Difference Index	$NIR - R + G$
GRVI	Green Ratio Vegetation Index	NIR/G

Table 7. Summary of color vegetation indices derived from the UAV-RGB ortho-image.

VI	Name	Formula
VARI	Visible Atmospherically Resistant Index	$(g - r)/(g + r - b)$
ExR	Excess Red Vegetation Index	$1.4r - g$
ExB	Excess Blue Vegetation Index	$1.4b - g$
ExG	Excess Green Vegetation Index	$2g - r - b$
ExGR	Excess Green minus Excess Red Vegetation Index	$ExG - ExR$
MGRVI	Modified Green Red Vegetation Index	$(g^2 - r^2)/(g^2 + r^2)$
WI	Woebbecke Index	$(g - b)/(r - g)$
IKAW	Kawashima Index	$(r - b)/(r + b)$
GLA	Green Leaf Algorithm	$(2g - r - b)/(2g + r + b)$
RGBVI	Red Green Blue Vegetation Index	$(g^2 - b \times r)/(g^2 + b \times r)$
GRVI	Green Red Vegetation Index	$(G - R)/(G + R)$
CIVE	Color Index of Vegetation	$0.441r - 0.881g + 0.385b + 18.78745$
COM	Combination	$0.25ExG + 0.3ExGR + 0.33CIVE + 0.12VEG$
GRRi	Green Red Ratio Index	G/R
VEG	Vegetative	$g/r \times 0.667 \times b \times 0.333$
GBRI	Green Blue Ratio Index	G/B
RBRI	Red Blue Ratio Index	R/B
INT	Color Intensity Index	$(R + G + B)/3$
IPCA	Principal Component Analysis Index	$0.994 \times R - B + 0.961 \times G - B + 0.914 \times G - R $

Later in the analysis, Pearson's correlation analysis was used to establish the relationships between spectral variables, Sentinel-2 10 m spectral variables, and the calibrated GEDI-L4A AGB at the plot level.

2.3. Methodology

2.3.1. Plot-Based AGB Estimation and Calibration Factor Extraction Using UAV-LiDAR Datasets

In this study, the first goal was to estimate AGB at the plot level. The UAV-LiDAR datasets extracted critical forest metrics for each plot, including tree height, crown diameter, leaf area index, gap fraction, and canopy cover statistics. Then, it was followed by the computation of DBH for every tree in the plot. As noted in prior studies, *Pinus massoniana* was the leading tree species in the studied area [64,76]. To estimate DBH, we used a general DBH formula (Equation (1)) using the coefficients ($a = 0.5996$, $b = 1.6509$, $c = 0.3204$) previously published for each tree species by [64]. After calculating the DBH for all trees within a plot, the mean DBH was determined to represent the plot level DBH. For each tree, the DBH was estimated as follows:

$$DBH = 0.5966 \times (CD^{1.6509}) \times (H^{0.3294}) \quad (1)$$

where CD = Crown Diameter, H = Tree Height

To estimate AGB for each tree, we applied the allometric equation for *Pinus massoniana*, as described by [76] (Equation (2)). This equation relates AGB to DBH and tree height. The resulting AGB was calculated in kilograms (kg) and converted to metric tons (Mg) by dividing by 1000.

$$AGBMg = (0.0487 \times (DBH^2 \times Height)^{0.932})/1000 \quad (2)$$

As the size of the plot is 25 m $\times$ 25 m (0.0625 hectares), the calculated AGB was further converted into Mg/ha by diving by 0.0625 as given in Equation (3):

$$AGBMg/ha = AGB/0.0625 \quad (3)$$

After estimating AGB for each plot, the values were compiled to enable correlation and calibration factor extraction for GEDI-L4A footprint AGB.

The UAV-LiDAR data were acquired for 60 plots at three sites, and AGB was estimated using the equations (Equations (1)–(3)). These estimates of AGB from UAV-LiDAR were compared to the GEDI-L4A AGB values and GEDI-L2A height values at a plot level. The purpose was to retrieve a calibration factor to bring the GEDI-L4A AGB and GEDI-L2A rh98 height values to provide better accuracy in the AGB prediction model.

2.3.2. Model Development Using Regression Technique

A random forest model (RF) is among the most utilized regression models employed by many studies for estimating forest AGB, along with XGBoost and SVM [10]. It is an integrated learning method that combines several decision trees to increase the accuracy and reliability of the regression tasks [77]. Random forest regression makes the model more robust against overfitting and better captures the complex interactions between predictors. The memory usage of this model, especially with big datasets, can be a limitation in a resource-constrained environment since each tree is constructed independently and kept in memory. XGBoost is the gradient-boosting algorithm applied in regression and classification; however, poorly tuned algorithms cause overfitting. Besides being computationally expensive to run, big data may also run slow since it is demanding for memory and processors. Indeed, Support Vector Machines (SVM) are a compelling model both for classification and regression problems but, at the same time, computationally run slow, especially when big datasets are in view.

In this study, the random forest (RF) model was selected and employed as the representative model. In addition, we compared RF with other models (XGBoost and SVM) for predicting AGB in the study area and evaluated the models using key performance metrics such as R^2 , RMSE, and MAE. Based on these analyses, the random forest (RF) model was selected as the most reliable for AGB prediction in the study area due to its strong performance across the evaluation metrics and its ecologically valid AGB predictions, as mentioned in Section 3.3, as shown in Table 8.

Table 8. Comparison of machine learning (ML) models' performance based on evaluation metrics.

ML Models	Evaluation Metrics	Model 1	Model 2	Comments
Random Forest	Training	$R^2 = 0.92$, RMSE = 15.9, MAE = 9.9	$R^2 = 0.94$, RMSE = 13.5, MAE = 8.1	Random forest yields excellent performance across all datasets, with high R^2 and relatively low RMSE and MAE, showing great generalization. It is thus the most robust model overall since it consistently results in training, testing, and validation, outperforming XGBoost and SVM models.
	Testing	$R^2 = 0.74$, RMSE = 24.9, MAE = 16.2	$R^2 = 0.75$, RMSE = 24.5, MAE = 15.7	
	Validation	$R^2 = 0.76$, RMSE = 25.6, MAE = 17.2	$R^2 = 0.78$, RMSE = 24.3, MAE = 16.5	
XGBoost	Training	$R^2 = 0.94$, RMSE = 12.4, MAE = 9.4	$R^2 = 0.91$, RMSE = 15.07, MAE = 11.2	XGBoost performs excellently in training but struggles with generalization to testing and validation, as indicated by its lower R^2 and higher error metrics than random forest. It is prone to overfitting, particularly in Model 1, which limits its reliability.
	Testing	$R^2 = 0.72$, RMSE = 25.3, MAE = 16.9	$R^2 = 0.81$, RMSE = 22.32, MAE = 15.6	
	Validation	$R^2 = 0.74$, RMSE = 26.1, MAE = 17.7	$R^2 = 0.75$, RMSE = 26.1, MAE = 16.7	
SVM	Training	$R^2 = 0.652$, RMSE = 39.67, MAE = 25.52	$R^2 = 0.75$, RMSE = 27, MAE = 16.6	SVM consistently shows average performance across all phases, with low R^2 and high RMSE and MAE values. This indicates weak generalization and limited effectiveness for both Model 1 and Model 2. It is excluded from consideration due to its inferior performance compared to random forest and XGBoost.
	Testing	$R^2 = 0.68$, RMSE = 32, MAE = 23.9	$R^2 = 0.72$, RMSE = 27.7, MAE = 16.5	
	Validation	$R^2 = 0.60$, RMSE = 38.9, MAE = 26.23	$R^2 = 0.65$, RMSE = 30.5, MAE = 16.7	

The RF model was applied to build two AGB prediction models, which integrated various data sources, including GEDI-L4A calibrated AGB, GEDI-L2A rh98 calibrated heights, Sentinel-2 spectral data, terrain features from NASADEM (such as slope and elevation), and UAV datasets spectral variables. These models were developed using the

Spyder application within Anaconda Navigator and trained on consistent datasets for each predictor and target variable (Table 9).

Table 9. Random forest regression model training hyperparameters.

Random Forest Model	Model Parameter	Value
RF-First Model (GED-L4A calibrated AGB (target variable) and GEDI-L2A (calibrated height), Sentinel-2 indices, and slope and elevation from NASADEM (predictor variable))	Bootstrap	True
	Sampling Method	10-fold cross-validation
	Type	Regression
	Max_depth	15
	Min_samples_leaf	4
	Min_samples_split	3
RF-Second Model (GED-L4A calibrated AGB (target variable) and GEDI-L2A (calibrated height), UAV-spectral variables (multispectral and RGB), canopy cover (UAV-LiDAR Forest metrics), Sentinel-2 Indices, and slope and elevation from NASADEM (predictor variable))	n_estimators	500
	Bootstrap	True
	Sampling Method	10-fold cross-validation
	Type	Regression
	Max_depth	20
	Min_samples_leaf	3
	Min_samples_split	2
	n_estimators	500

The first model estimates AGB using a random forest regression approach. The dependent variable is the calibrated GEDI-L4A mean AGB, while predictable variables include GEDI-L2A rh98 heights, slope and elevation derived from NASADEM, and Sentinel-2 spectral indices (Tables 3 and 9). All data are reprojected to a standard 10 m spatial resolution, and Sentinel-2 vegetation indices are calculated from 10 m bands. First, the rh98 calibrated height from GEDI-L2A was interpolated to the exact resolution using the inverse distance weighting (IDW) method.

The second model is more general for the prediction of AGB at a district scale and combines the UAV spectral data with the calibrated heights from the GEDI-L2A, slope and elevation data, canopy cover (CC) from UAV LiDAR Forest metrics, as well as the spectral indices from Sentinel-2, as in Table 3. A correlation analysis between spectral variables from UAV-multispectral and RGB datasets and the calibrated AGB was performed, as discussed in Section 3.2. Both models underwent hyperparameter tuning through grid search and 10-fold cross-validation to identify the best parameters, as summarized in Table 9. The dataset was partitioned into 70% for training/testing and 30% for validation to ensure robust model evaluation and generalization of model performance. After training, the models predicted AGB across the district and calculated prediction uncertainty by estimating standard errors. The model's performance was evaluated based on the coefficient of Determination (R^2), Mean Absolute Error (MAE), and Root Mean Square Error (RMSE).

2.3.3. Model Accuracy Assessment

While presenting both models, the performance of the random forest regression model regarding the inter-relationship of target and predictor variables was gauged along with accuracy through R^2 , RMSE, and MAE metrics. In addition, the scatter plots were plotted for cross-validation, reflecting how consistently the result came up from the model. The R^2 gave the proportion of variance in the observed AGB explained by the model. The RMSE provided an idea of the average magnitude of the prediction errors, with more significant errors weighted more. A low RMSE indicated that the RF model produced predictions close to the observed values, indicating high accuracy. At the same time, MAE showed the average of the absolute errors, a simple, robust degree of prediction accuracy that treats all errors equally.

3. Results

3.1. Calibration Factor Extraction Using UAV-LiDAR Datasets

A linear regression analysis correlated GEDI-L2A rh98 height values against UAV-LiDAR-derived mean heights. Outlier plot values were removed to improve the accuracy of this assessment further. After outlier removal, 37 plots remained in the analysis, and the correlation improved to $R^2 = 0.9722$. The overall calibration equation to correct GEDI-L2A height values was derived in Equation (4). The performed analysis is graphically shown in the scatter plot of Figure 4, which evidences the relationship between GEDI-L2A rh98 height values and UAV-LiDAR-derived mean heights at plot level.

$$y = 1.1346x \quad (4)$$

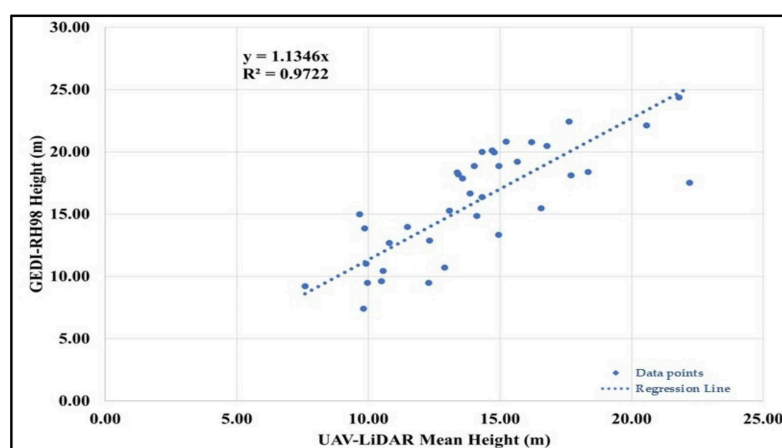


Figure 4. Scatter plot comparing GEDI-L2A rh98 heights and UAV-LiDAR mean heights.

Linear regression assessed the relationship between UAV-LiDAR-derived AGB (as discussed in Section 2.3.1) and GEDI-L4A AGB values. The first iteration yielded a coefficient of correlation (R^2) is 0.78 with equation $y = 0.6198x$ with zero intercept. This was reconsidered as the outlier plot values were removed, after which the analysis was performed with 46 plots that significantly improved the correlation of $R^2 = 0.9187$. The final calibration equation for the correcting GEDI-L4A AGB values is given in Equation (5). This analysis is illustrated in the scatter plot mentioned in Figure 5.

$$y = 0.7499x \quad (5)$$

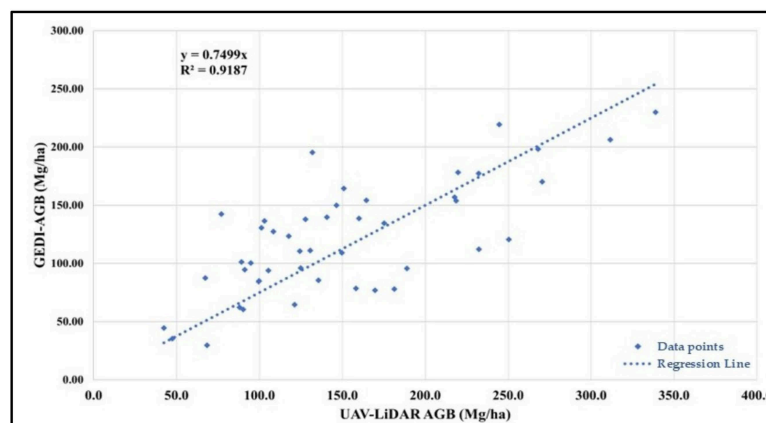


Figure 5. Scatter plot comparing GEDI-AGB and UAV-LiDAR AGB in Mg/ha.

Following the earlier analysis, the calibration factors were used to calibrate the GEDI-L4A AGB and GEDI-L2A height values using Equations (4) and (5), respectively. Later, these calibrated values of AGB and height were considered the target and predictor variables for the RF prediction model, respectively, mentioned in Table 9. Figure 6 shows the calibrated GEDI footprints ($n = 2096$) containing AGB and height values.

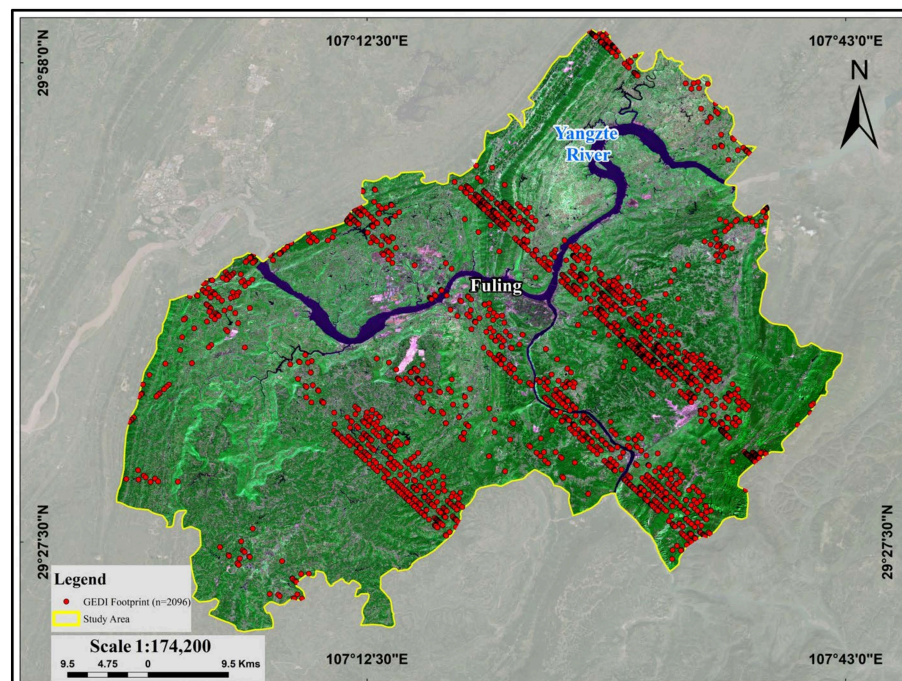


Figure 6. Calibrated GEDI footprints for AGB and height values in Fuling District.

3.2. Correlation Analysis Between Spectral Variables from UAV and Sentinel-2 Datasets

The Pearson's correlation coefficients (r) between calibrated AGB and spectral signatures were estimated using the data from 17 spectral variables comprising 13 vegetation indices and four multispectral channels derived from UAV-multispectral imagery (Figure 7). For the data derived from UAV-RGB imagery, the correlation analysis was calculated for 22 spectral variables (19 vegetation indices and three RGB channels) to explore the relationship between calibrated AGB and spectral signatures (Figure 8). The analysis was computed for 13 spectral variables comprising nine vegetation indices and four spectral (10 m resolution) from Sentinel-2 imagery (Figure 9). These correlations were conducted to investigate the association of calibrated AGB with all datasets' spectral signatures. Moreover, the more correlated spectral variables were used in the linear regression of over 60 surveyed plots, the more contributing spectral variables were identified. To maintain each dataset's spatial alignment with the Sentinel-2 spectral variables, each selected variable of UAV-derived spectral variables was extrapolated to the Fuling District boundary using linear regression at the 10 m resolution and included as predictor variables in the second model of AGB estimation, as mentioned in Table 9.

The correlation results of the spectral variable of the UAV-multispectral dataset indicate that most of the spectral variables are highly interrelated (Figure 7). Specifically, eight variables, which are the Chlorophyll Index-red edge (CI_{re}), Chlorophyll Vegetation Index (CVI), Green Normalized Vegetation Index (GNDVI), Modified Simple Ratio-red-edge (MSE-re), Normalized Difference Vegetation Index (NDVI), Red-Edge Normalized Difference Vegetation Index (NDVI_{re}), Green Ratio Vegetation Index (GRVI), and Simple Ratio-red-edge (SR_{re}), showed highly positive values of correlations, reaching up to 1. All these variables were highly correlated with the other studied variables, except for their av-

erage correlation with the Green Difference Index (GDI). Furthermore, the Green Difference Index (GDI) still yielded good correlations with the Chlorophyll Vegetation Index (CVI), Green Chlorophyll Index (CIgreen), and Red-Edge Triangular Vegetation Index (RTVcore). Considering multispectral channels, the near-infrared (NIR) bands indicated moderate correlations with the Chlorophyll Vegetation Index (CVI), Red-Edge Triangular Vegetation Index (RTVcore), and Green Difference Index (GDI). Similarly, red-edge bands showed moderate correlations with the Chlorophyll Vegetation Index (CVI), Red-Edge Triangular Vegetation Index (RTVcore), Green Difference Index (GDI), and Green Ratio Vegetation Index (GRVI). In contrast, both red and green channels showed moderate correlations with the Chlorophyll Vegetation Index (CVI), Red-Edge Triangular Vegetation Index (RTVcore), and Green Difference Index (GDI). While spectral information did vary, most of the variables calculated from UAV-based multispectral imagery showed moderate to weak positive relations to calibrated AGB. The highest correlation coefficient (r) between spectral variables and calibrated AGB was found for GRVI and CIgreen ($r = 0.43$) and MSR ($r = 0.39$). These results show that all 17 spectral variables could be used as predictor variables in the AGB prediction model to ensure its accuracy and realization of good predictability.

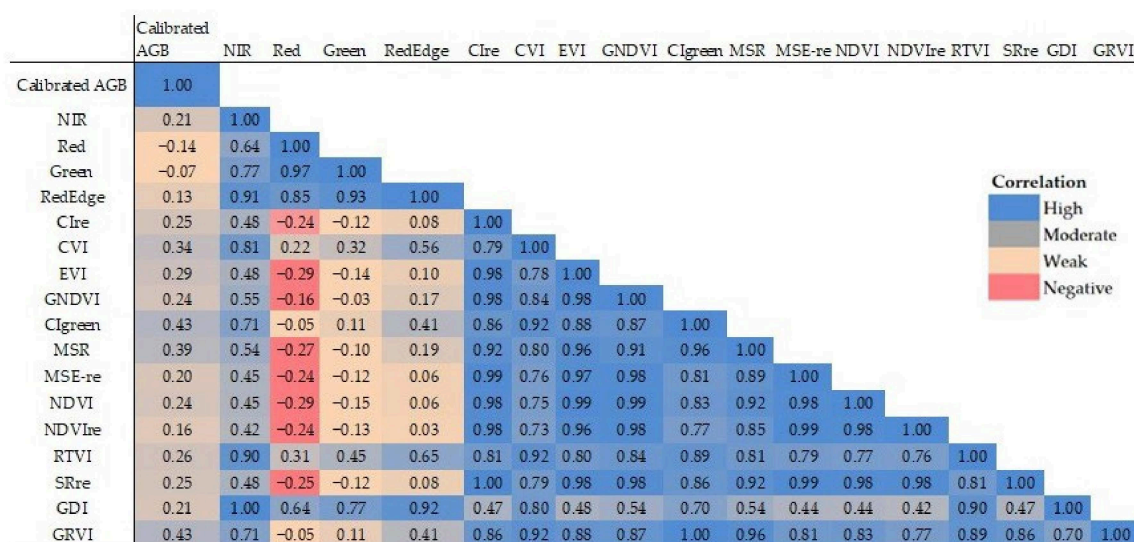


Figure 7. Pearson's correlation analysis between UAV-multispectral spectral variables and the calibrated GEDI AGB.

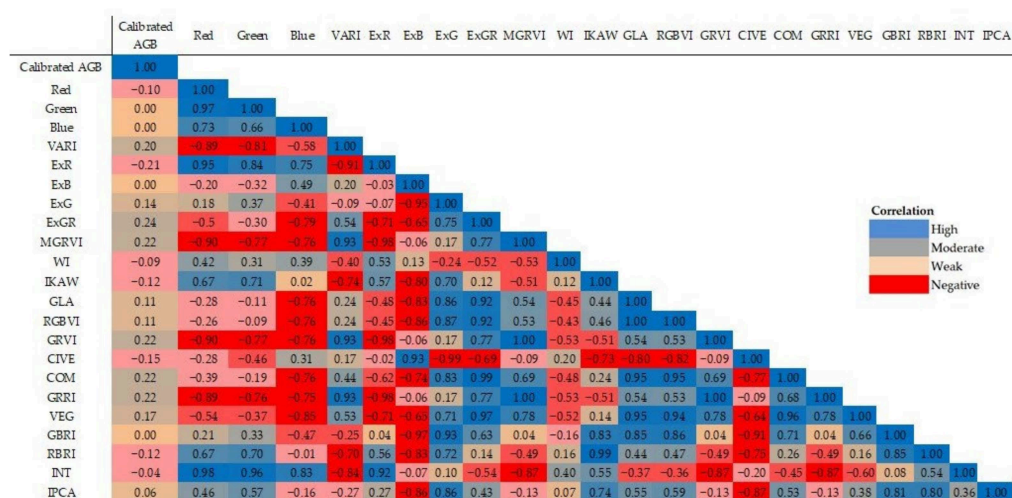


Figure 8. Pearson's correlation analysis between UAV-RGB spectral variables and the calibrated GEDI AGB.

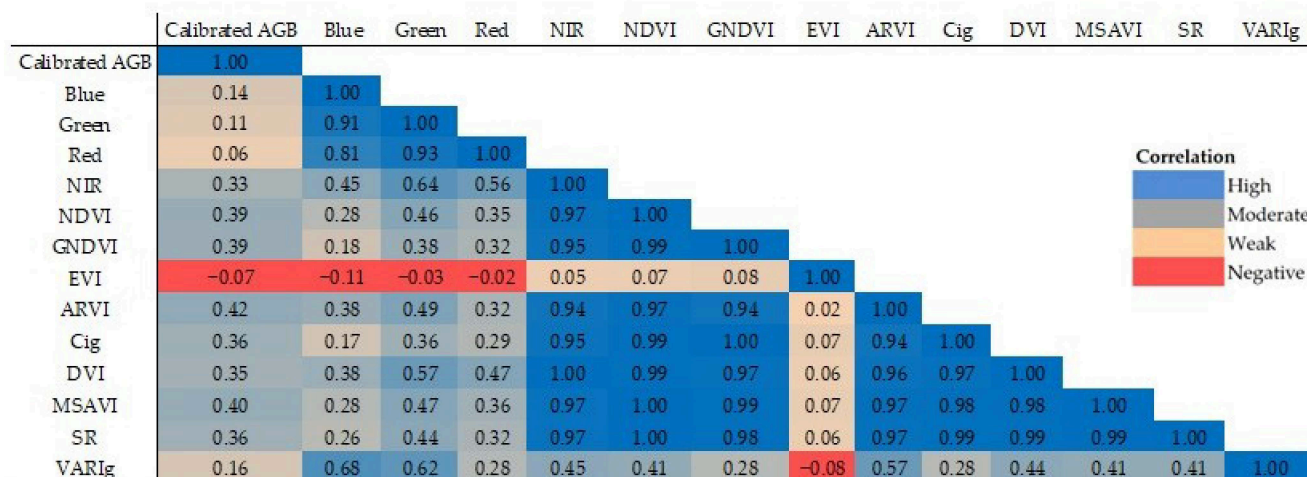


Figure 9. Pearson's correlation analysis between Sentinel-2 imagery-derived spectral variables and the calibrated GEDI AGB.

The correlation was moderate in most of the spectral variables of UAV-RGB imagery (Figure 8). Highly positive significant correlations were found among seven variables: Excess Green Vegetation Index (ExG), Excess Green minus Excess Red Vegetation Index (ExGR), Modified Green Red Vegetation Index (MGRVI), Kawashima Index (IKAW), Red Green Blue Vegetation Index (RGBVI), Green Red Vegetation Index (GRVI), and Green Blue Ratio Index (GBRI), having extreme values up to 1. Some of the variables showed negative correlations, such as Excess Red Vegetation Index (ExR), Excess Blue Vegetation Index (ExB), Woebbecke Index (WI), Color Index of Vegetation (CIVE), and Color Intensity Index (INT). In terms of RGB channels, both red and green bands exhibited moderate correlations with the Kawashima Index (IKAW), Green Blue Ratio Index (GBRI), Red Blue Ratio Index (RBRI), and Principal Component Analysis Index (IPCA). Despite spectral variations, most variables from UAV-based RGB imagery showed weak positive correlations with calibrated AGB. The highest correlation coefficient (r) with calibrated AGB was found for the Modified Green Red Vegetation Index (MGRVI), Green Red Vegetation Index (GRVI), Combination (COM), and Green Red Ratio Index (GRRI) ($r = 0.22$). Accordingly, all variables possessing negative correlations, including ExR, ExB, WI, CIVE, and INT, were excluded based on this analysis. Thus, 17 variables, consisting of 14 vegetation indices and three RGB channels) could be used as predictor variables in the AGB prediction model.

In most spectral variables, using Sentinel-2 data generally resulted in high positive correlations, except for EVI (Figure 9). The NIR bands were more strongly related to all vegetation indices within all channels. Most calibrated AGB correlated moderately positively with most variables derived from Sentinel-2 imagery. The highest correlation coefficient was found for Atmospherically Resistant Vegetation Index (ARVI) ($r = 0.43$), Modified Soil Adjusted Vegetation Index (MSAVI) ($r = 0.41$), Normalized Difference Vegetation Index (NDVI), and Green Normalized Difference Vegetation Index (GNDVI) ($r = 0.40$). Based on the above analysis, the Enhanced Vegetation Index (EVI) was excluded because it had a weaker correlation. Finally, to enhance their performance and predictive accuracy, 12 spectral variables comprising eight vegetation indices and four spectral channels were chosen to apply in both RF AGB prediction models (Table 9).

The final spectral variable from UAV datasets selected among the shortlisted 17 UAV-multispectral and 14 UAV-RGB variables, as discussed in Section 3.2, are included as predictor variables in the second AGB RF prediction model (Section 3.3.2 and Table 9). The variables are six from UAV-multispectral datasets, namely, NDVIreMS, CIVIMS, GreenMS, RededgeMS, MSRreMS, and NDVIMS; six UAV-RGB spectral variables, namely, RBRIRGB,

IPCARGB, IKAWRGB, GreenRGB, RedRGB, and BlueRGB; and one from UAV-LiDAR forest metrics: canopy cover (CC).

3.3. AGB Estimation Model Development

3.3.1. Estimation of AGB Using Calibrated GEDI Datasets and Sentinel-2 Indices: First Model

AGB was predicted using the calibrated GEDI-L4A AGB and GEDI-L2A height dataset and the first RF regression model for the Fuling District, described in Table 9. The spatial extent of the predicted AGB and its uncertainty map (in the form of a standard error of the prediction) are presented in Figure 10. The target variable in the model was calibrated GEDI-L4A AGB. In contrast, predictable variables include GEDI-L2A rh98 calibrated height data, slope and elevation derived from NASADEM, and Sentinel-2 spectral indices (detailed in Tables 3 and 8). All datasets were reprojected to a standard 10 m spatial resolution, and Sentinel-2 vegetation indices were calculated from 10 m bands. The calibrated height data from GEDI-L2A were interpolated to match this resolution using the IDW method.

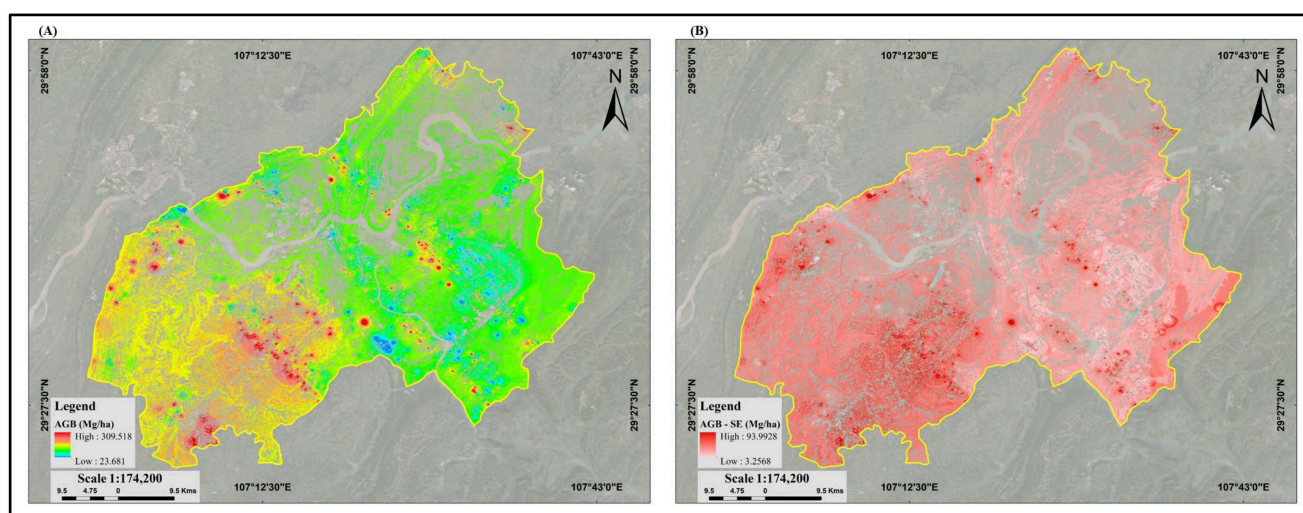


Figure 10. First RF model prediction. (A) AGB prediction in Mg/ha. (B) Standard error of AGB prediction in Mg/ha.

Hyperparameter tuning was performed by grid search along with 10-fold cross-validation to find the best parameters, as summarized in Table 9. The data were partitioned into two parts: 70% for training and testing and 30% for validation to obtain a substantial generalization of model performance. Results were depicted visually by the scatter plots of training, testing, and validation data, showing the correlation between GEDI-calibrated AGB and predicted AGB in Mg/ha, represented in Figure 11A–C. The model assessment statistics were indicative of a strong predictive performance based on the training data, with the coefficient of determination (R^2), root mean square error (RMSE), and mean absolute error (MAE) values as follows. For training data: $R^2 = 0.917$, RMSE = 15.90 Mg/ha, MAE = 9.94 Mg/ha; for testing data: $R^2 = 0.742$, RMSE = 24.94 Mg/ha, MAE = 16.18 Mg/ha; and for validation data: $R^2 = 0.758$, RMSE = 25.63 Mg/ha, MAE = 17.22 Mg/ha, respectively (Table 9). These variables include calibrated height (H_Cal_10), VARlg, slope, elevation, DVI, ARVI, and GNDVI, which the model showed as more critical, as shown in Figure 11D.

The final prediction of this model was exported in tiff format at 10 m resolution. Then, the forest mask layer of 2022 [25] was used to mask the AGB prediction and its standard error over Fuling District. The study area's minimum and maximum AGB predictions ranged from 23.68 Mg/ha to 309.5 Mg/ha, and the standard error was 3.25 Mg/ha to 93.9 Mg/ha, as shown in Figure 10.

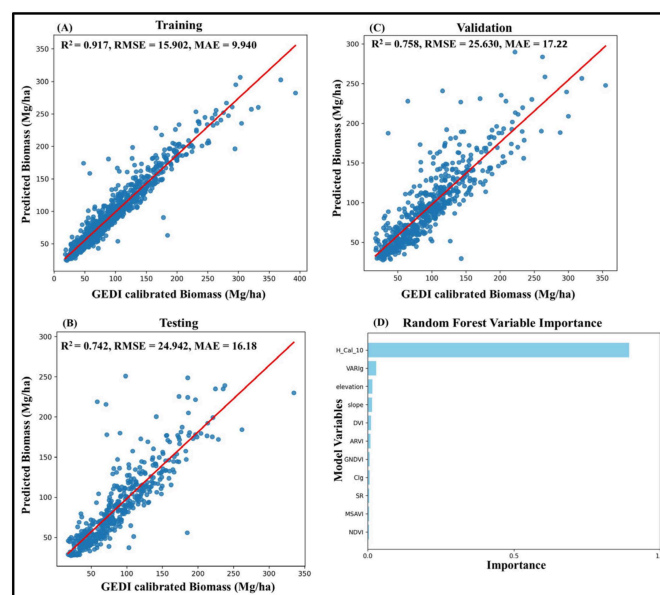


Figure 11. Model assessment statistics of the AGB prediction using calibrated GEDI-L4A AGB and GEDI-L2A heights for Fuling District. (A) Scatter plot of training data. (B) Scatter plot of testing data. (C) Scatter plot of validation data. (D) Variable importance (relative significance indicators within the model). Note: Relatively low significance indicators in (D): GNDVI, Cig, SR, MSAVI, and NDVI.

3.3.2. Estimation of AGB Using Calibrated GEDI, Sentinel-2 Indices, and Spectral Variables from UAV-Multispectral and RGB: Second Model

AGB was predicted using the calibrated GEDI-L4A AGB and GEDI-L2A height dataset and the second RF regression model for the Fuling District, described in Table 9. The spatial extent of the predicted AGB and its uncertainty map (in the form of a standard error of the prediction) is presented in Figure 12. The target variable in the model was calibrated GEDI-L4A mean AGB. In contrast, predictable variables include GEDI-L2A rh98 calibrated height data, spectral variables from UAV-multispectral and RGB, slope and elevation derived from NASADEM, and Sentinel-2 spectral indices (detailed in Tables 3 and 9). The UAV-datasets-derived indices were shortlisted, as discussed in Section 3.2. All datasets were reprojected to a standard 10 m spatial resolution, and Sentinel-2 vegetation indices were calculated from 10 m bands. The calibrated height data from GEDI-L2A was interpolated using the IDW method at 10 m resolution to match this resolution.

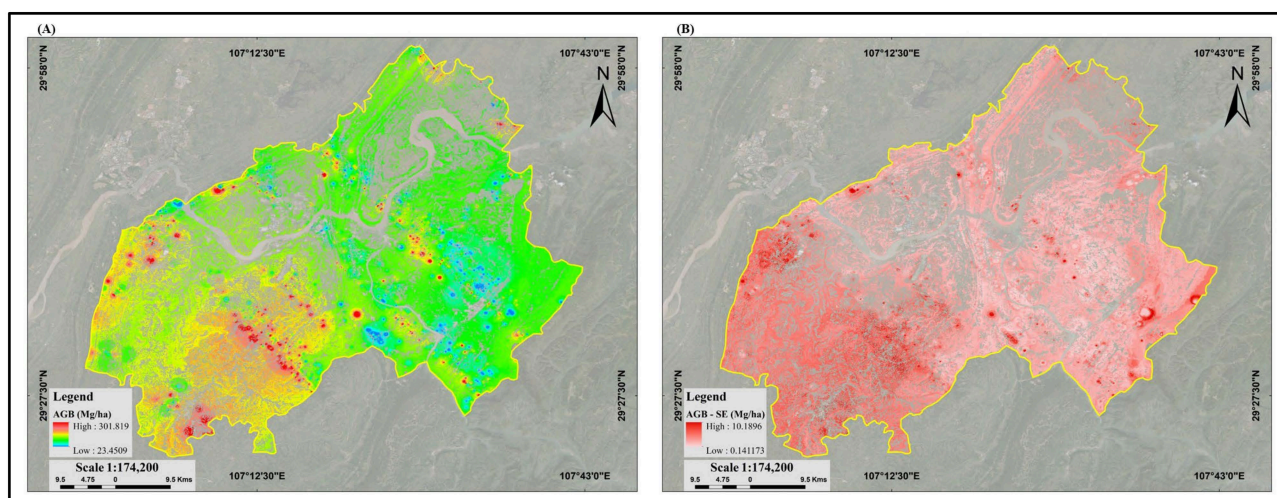


Figure 12. Second RF model prediction. (A) AGB prediction in Mg/ha. (B) Standard error of AGB prediction in Mg/ha.

Hyperparameter tuning was performed by grid search along with 10-fold cross-validation to find the optimum parameters, as summarized in Table 9. The data were partitioned into two parts: 70% for training and testing and 30% for validation to obtain a substantial generalization of model performance. Results were depicted visually by the scatter plots of training, testing, and validation data, showing the correlation between GEDI-calibrated AGB. They predicted AGB in Mg/ha, represented in Figure 13A–C. The model assessment statistics were indicative of a strong predictive performance based on the training data, with the R^2 , RMSE, and MAE values as follows. For training data: $R^2 = 0.941$, RMSE = 13.514 Mg/ha, MAE = 8.136 Mg/ha; for testing data: $R^2 = 0.75$, RMSE = 24.48 Mg/ha, MAE = 15.70 Mg/ha; and for validation data: $R^2 = 0.781$, RMSE = 24.37 Mg/ha, MAE = 16.54 M, respectively (Table 9). These variables include calibrated height (H_Cal_10), VARlg, slope, elevation, RBRIRGB, DVI, CC, ARVI, NDVIreMS, and CIVIMS, which the model showed to be more critical, as shown in Figure 13D.

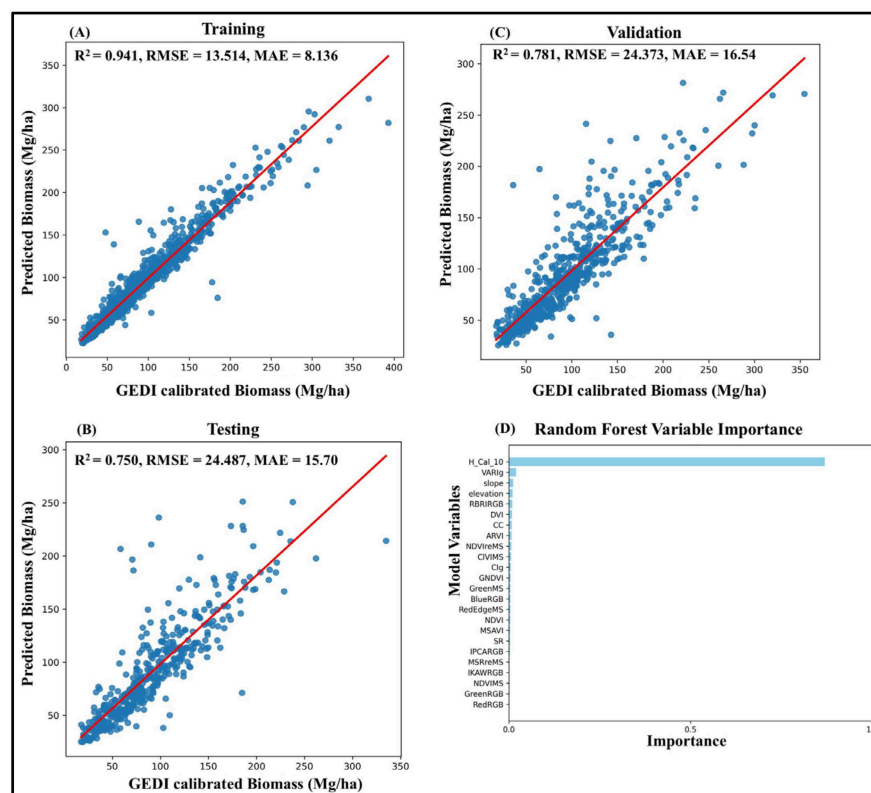


Figure 13. Model assessment statistics of the AGB prediction using calibrated GEDI-L4A AGB, GEDI-L2A heights, and spectral variables from UAV-multispectral and RGB datasets for the Fuling District. (A) Scatter plot of training data. (B) Scatter plot of testing data. (C) Scatter plot of validation data. (D) Variable importance (relative significant parameters within the model). Note: Relatively low significance parameters in (D): Sentinel-2 (CIg, GNDVI, NDVI, MSAVI, SR), UAV-multispectral (Green, MSRre, NDVI), UAV-RGB (Green, Red, IPCA, IKAW).

The final prediction of this second model was exported in tiff format at 10 m resolution. Then, the forest mask layer of 2022 [25] was used to mask the AGB prediction and its standard error over Fuling District. The study area's minimum and maximum AGB predictions ranged from 23.45 Mg/ha to 301.81 Mg/ha, and the standard error was 0.14 Mg/ha to 10.18 Mg/ha, as shown in Figure 12.

3.3.3. Comparative Study of Machine Learning Models

The random forest (RF) model was selected and employed as the representative model in this study. In addition, we compared RF with other models (XGBoost and SVM) for

predicting AGB and evaluated the models using key performance metrics such as R^2 , RMSE, and MAE. Based on this analysis, SVM was excluded from further analysis due to its lower performance compared to the other models, particularly regarding R^2 , RMSE, and MAE, making it less suitable for this study. XGBoost showed a higher R^2 (0.94) but struggled with generalization to testing and validation, as indicated by its lower R^2 and higher error metrics (RMSE and MAE) than random forest. It is prone to overfitting, particularly in Model 1, which limits its reliability. We decided to exclude XGBoost from further analysis despite its higher R^2 in training for Model 1 only. As a result, random forest (RF) was the most reliable model for AGB prediction in the study area for both models due to its strong performance across the evaluation metrics and its ecologically valid AGB predictions aligned with other literature [20] in a study area, as shown in Table 8.

4. Discussion

This research presents a novel method for estimating AGB at the district level by calibrated GEDI-L4A AGB values using UAV-LiDAR ALS heights, Sentinel-2 spectral indices, and integrating spectral variables from UAV-multispectral and RGB datasets to develop machine learning models specific to the study area. In this respect, due to the absence of ground-truth data, especially on DBH, we restricted our study to only UAV-LiDAR ALS data that comprises height and crown diameter measurements. Previous studies [78,79] have shown that including crown parameters, such as crown area, diameter, and height, can significantly enhance AGB estimation accuracy. These studies identified limitations that could affect accuracies, such as inconsistent trajectories, variable image quality across overlapping images, the inherent nature of instrument measurement subject to systematic errors, and the manual data extraction of the trees, accomplished using handheld GPS devices. In the study area, other studies [64,76] have developed generic allometric equations and coefficients for *Pinus massoniana* in the Chongqing region with height and DBH relationship. The AGB estimation in this study was generalized using these allometric equations only for this region. We used plot-level AGB estimates from 60 surveyed plots to derive a calibration factor through correlation with GEDI-L4A AGB and GEDI-L2A rh98 height values mentioned in Sections 2.3.1 and 3.1. The results show that, after removing outliers, 46 plots yielded an R^2 of 0.918 for AGB estimates, while 37 plots for GEDI-L2A rh98 heights had an impressive R^2 of 0.97. These plots were instrumental in extracting a calibration factor that allowed us to normalize the unsurveyed GEDI AGB and height footprint values, as illustrated in Figure 6.

We analyzed the importance of spectral variables from UAV-multispectral and UAV-RGB datasets using Pearson correlation analysis to identify the most critical spectral variables that improve the accuracy and robustness of AGB prediction models. Previous studies have used UAV-multispectral indices [39,80] and UAV-RGB ([37,81] spectral variables' assessment and evaluations to integrate with the prediction model to enhance the AGB estimates. Some limitations revolving around these studies on UAV-based multispectral imagery concern poor effectiveness in unsupervised classifications, satellite-scale canopy height data, species-specific data availability, and environmental condition variability, which result in limitations in spatial resolution. This has implications for result accuracy and applicability. The limitations among the studies dealing with UAV-RGB imagery concern the lack of evaluation of texture feature extraction across multiple spatial resolutions and constraints in the AGB estimation of vegetation indices without the NIR band. In general, the inability of the study model to characterize local spectral morphological variability reduces its overall explanatory power. Variability within revisit time and inconsistencies in the lighting and sensor make other differences and affect result accuracy. This has been successful in our study that focused on the most pertinent spectral

variables from UAV-multispectral, UAV-RGB, and Sentinel-2 datasets, including the canopy cover extracted from the UAV-LiDAR data. The correlation analysis has shown that this is a practical approach to detecting variation in spectral variables' calibrated GEDI AGB and among the variables, as shown in Figures 7–9. After a closer look into these most critical variables, a final correlation assessment with calibrated GEDI AGB using six main UAV-multispectral, six main UAV-RGB, and one UAV-LiDAR forest metric spectral variable was performed. This was finally used to present the AGB prediction model, as outlined in Section 3.2, whose important variables are shown in Figures 10 and 12.

For developing the RF regression for the first model, calibrated GEDI-L4A AGB and GEDI-L2A rh98 height data were combined with Sentinel-2 spectral variables and slope and elevation from NASADEM. In contrast, spectral variables from UAV-multispectral and RGB were added to the datasets to estimate AGB in the Fuling District for the second model. Table 9 illustrates the datasets used for each model. Previous studies have successfully used spaceborne LiDAR datasets (GEDI products) with Sentinel-2 spectral variables and regression models to predict AGB and associated uncertainties [23,75,82]. Limitations of these studies include using land-cover information as a predictor instead of agro-climatic zone data that may improve model performance. In addition, water content in some species leads to skewed spectral features in the relationship, hence giving a stronger correlation of vegetation indices with dry weight AGB than with vegetation itself. These factors weaken the accuracy and robustness of models. In our study, the first model included the calibrated GEDI AGB as the dependent variable and was integrated with the calibrated height and Sentinel-2 indices, showing strong performance metrics of R^2 , RMSE, and MAE for training and testing in a random forest regression model. The second model, including the spectral variables from UAV-multispectral and RGB datasets and canopy cover data in addition to the above data, improved model metrics for training, testing, and validation, as shown in Figure 12. The AGB predictions in the Fuling District from the second model varied between 23.45 Mg/ha and 301.81 Mg/ha. The first model predicted 23.68 Mg/ha and 309.5 Mg/ha. Moreover, the second model's standard error (uncertainty map) showed a significant enhancement: values ranged from 0.14 Mg/ha to 10.18 Mg/ha, as shown in Figure 13. The results reveal that including spectral variables from UAV-multispectral and RGB datasets enhanced the normalization of AGB values and reduced the standard error. The regression model appeared capable of reproducing accurate AGB predictions using calibrated GEDI-L4A AGB and GEDI-L2A rh98 height values with Sentinel-2 and spectral variables from UAV-multispectral and RGB datasets. It finally provides a robust basis and framework for integrating the airborne and spaceborne datasets.

This study shows that calibrated AGB and height values normalized the AGB prediction model and the standard error by including spectral variables from UAV-multispectral and RGB datasets. In summary, these remote sensing workflows are foundational in integrating airborne UAV datasets that inform the calibration of spaceborne data and ultimately contribute to accurate and consistent AGB predictions at local and regional scales.

Limitations

In this study, 60 plots captured by the UAV platform were used for AGB estimation at the plot level, where a calibration factor was extracted to adjust the GEDI AGB and height values along with UAV spectral variables for the AGB model prediction. Increasing the number of plots would probably increase the accuracy and consistency of such a model, thus forming a more dependable basis for the AGB estimate. As mentioned in Section 2.2.1, GEDI-L4A AGB datasets were limited to timelines from 1 January 2022 to 31 December 2022. However, this limitation would be a drawback in understanding

long-term forest AGB dynamics. The temporal variability of AGB will be considered in the future.

5. Conclusions

This study assessed the Fuling District AGB using UAV and other remote sensing data. A calibration factor was proposed by relating the plot-level AGB estimates to the GEDI dataset to calibrate its AGB and height values. Pearson correlation analysis was undertaken to identify key spectral variables from UAV and Sentinel-2 datasets. The calibrated GEDI AGB, height values, and key spectral variables are integrated with a random forest model to estimate AGB in a study area. The key conclusions arising from this study are as follows:

- (1) The proposed calibration factor can effectively calibrate AGB and height information from GEDI-L4A and GEDI-L2A footprints.
- (2) Pearson correlation analysis identified VARIG, RBRI, DVI, CC, ARVI, NDVI_{re}, and CIVI as the key spectral variables for AGB estimation modeling.
- (3) RF regression models perform well when calibrated AGB and height data are combined and integrated with airborne (UAV) and spaceborne (Sentinel-2, slope and elevation from NASADEM) datasets.
- (4) Combining calibrated AGB and height values with spectral information from UAV data can improve the accuracy of AGB estimation.

This investigation reveals a hybrid approach to data fusion between the airborne UAV and spaceborne datasets from GEDI and Sentinel-2, which might advance AGB estimations. The approach provides more accurate AGB predictions and solves the uncertainties that have been a common scenario in global models. This could be very helpful in future global-scale AGB modeling, providing a strong backbone for calibrating global models. In the future, deeper integration of field plot data and inclusion of future mission data, such as BIOMASS and NISAR, would be practical. This reduces the uncertainties in AGB forest estimates and will improve the overall accuracy of AGB monitoring. The results derived from the present study provide a beneficial methodology that can be referred to in ecological research to guide sustainable forestry practices and inform environmental monitoring efforts.

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